

ENTERPRISE AI INFRASTRUCTURE

IOS+ Middleware Engine & COS+ Database

Infrastructure Intelligence for the Agentic Enterprise

Seven-Layer Ontology · Gate 530 Compliance · Evidence Fabric · RAG Vault



SMEPro Technologies

Investor Briefing — May 2026



Seven-Layer Ontology

Structured semantic routing



Gate 530 Compliance

Inference-time enforcement



Evidence Fabric

Immutable audit chain



RAG Vault Compliance-aware retrieval

The Enterprise AI Deployment Crisis



Most AI investments stall before they scale

87%



of enterprise AI pilots **fail to reach production**

GARTNER / MCKINSEY, 2024

\$2.4T



annual value **locked in siloed, untrustworthy AI systems**

IDC AI SPENDING FORECAST, 2025

14x



average middleware layers a **query traverses** before resolution
SMEPRO INTERNAL RESEARCH, 2025

0



purpose-built compliance-aware AI middleware platforms on the market
MARKET ANALYSIS, Q1 2026



Enterprises lack a **unified, compliance-aware middleware layer** that can govern, route, and evidence AI decisions across heterogeneous infrastructure — **until**

One Unified Intelligence Stack

IOS+ Middleware Engine orchestrates. COS+ Database remembers. Together they govern.



Gate 530 Logic

Compliance enforcement at every inference boundary — 127 dimensions evaluated in real time.



Seven-Layer Ontology

Structured semantic routing from raw data ingestion to governed insight delivery.



Evidence Fabric

Immutable audit trail for every AI decision — cryptographic chain of custody at output.



RAG Vault

Context-aware knowledge retrieval engine — compliance-screened retrieval across 2B+ document corpora.



YBR-L4 Architecture

Purpose-built for enterprise agentic workloads — not adapted from general-purpose infrastructure.



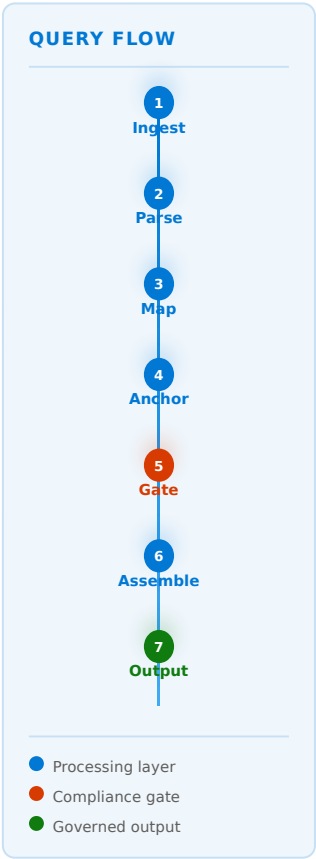
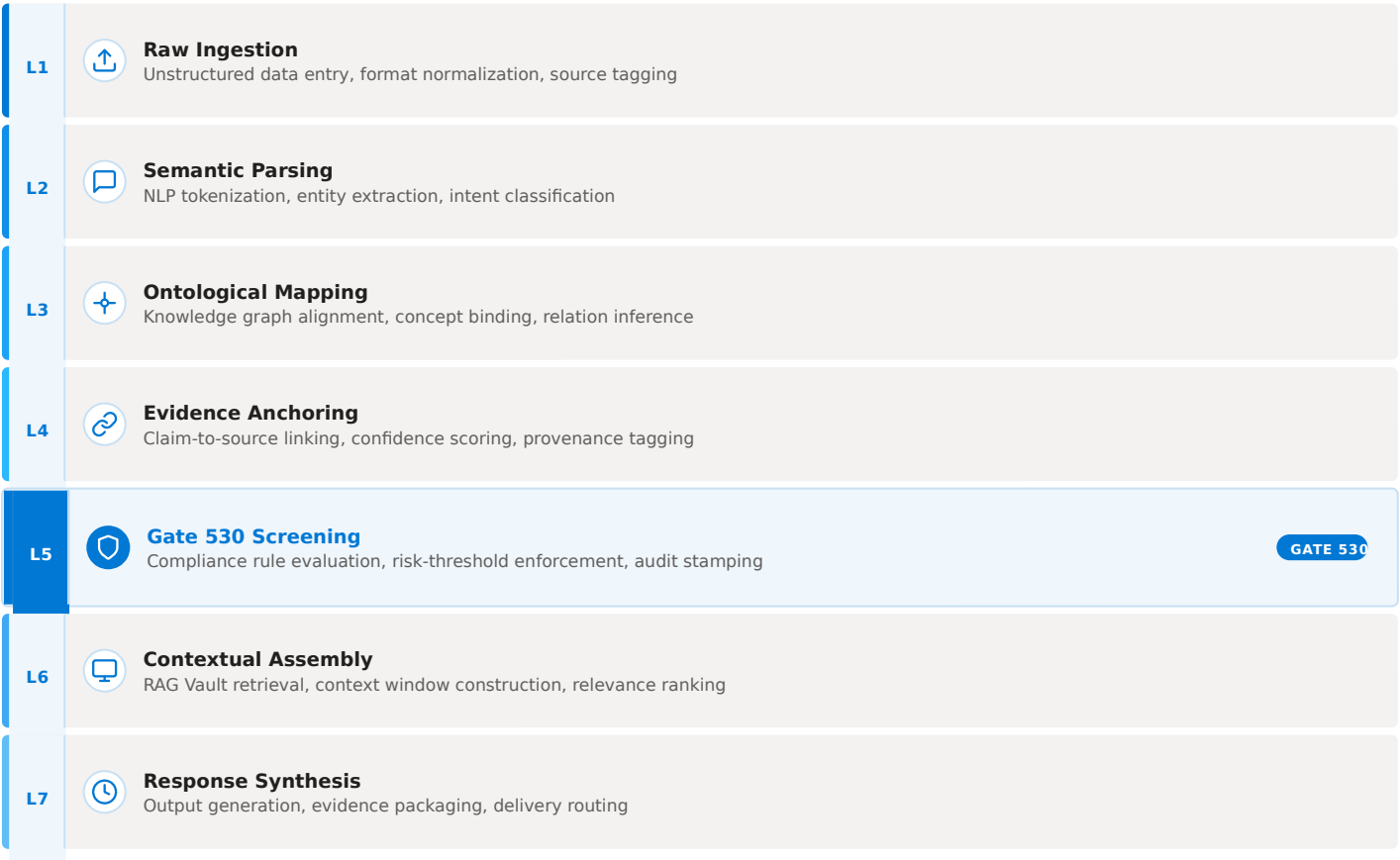
COS+ DB

Compliance-native columnar object store — compliance as the primary index dimension, not an afterthought.

Seven-Layer Ontology

The Architecture of Structured Intelligence

Every query traverses a governed semantic stack — no shortcuts, no blind spots



The only compliance engine native to the inference path — not bolted on after

⊗ Without Gate 530 LEGACY

- ⌚ Post-hoc compliance review adds 4-18 hour latency per inference batch — blocking real-time workflows
- 🔒 No real-time enforcement of data residency rules — violations propagate silently across inference calls
- 📄 Audit trails reconstructed from logs — incomplete by design, inadmissible under strict regulatory review
- ⚠️ Regulatory violations discovered weeks after exposure — remediation cost 10-50x higher than prevention

VS

✅ With Gate 530 IOS+ NATIVE

HIPAA SOC 2 GDPR CCPA FEDRAMP

- ⚡ Sub-50ms compliance decision at every inference step — zero workflow interruption, zero latency penalty
- 🛡️ Real-time HIPAA, SOC 2 & GDPR rule evaluation inline — 127 compliance dimensions per inference call
- 📝 Immutable decision record created at gate passage — cryptographically sealed, court-admissible audit trail
- ✅ Zero retroactive compliance — violations blocked at gate, not discovered in post-audit weeks later

Evidence **Fabric**

Every AI output arrives with a complete, immutable chain of custody



LAYER 04 · ORIGIN TRACKING

Immutable Provenance

Cryptographic linking of every output to its source documents, model version, retrieval context, and compliance gate stamp. No claim

without evidence.

✓ **Cryptographic Binding**



LAYER 04 · TRUST SCORING

Confidence Scoring

Dynamic confidence metric computed across source freshness, semantic distance, gate passage status, and retrieval relevance. Human-readable

trust signals.

✓ **Multi-Dimension Signal**



LAYER 04 · TIME-LOCK

Temporal Anchoring

Every inference record timestamped and version-locked. Regulatory reproducibility: any output can be reconstructed exactly as produced at

time of generation.

✓ **Regulatory Reproducibility**



Evidence Fabric turns AI from a black box into a fully auditable knowledge instrument — every decision traceable, every claim sourced, every record immutable.

Knowledge That Knows Its Place

A compliance-aware retrieval architecture purpose-built for regulated enterprise knowledge



Tiered Access Control

Knowledge partitioned by classification level. Retrieval requests screened through Gate 530 before context assembly.



Semantic Vector Search

Dense retrieval across structured and unstructured corpora with ontology-guided re-ranking at L3.



Live Knowledge Refresh

Continuous ingestion pipeline keeps RAG Vault current. No stale context served after TTL expiry.



Evidence-Linked Retrieval

Every chunk returned carries source metadata, gate stamp, and confidence band — directly feeding Evidence Fabric.

The Memory Layer Built for Compliance

A columnar object store engineered for compliance-native AI workloads — not adapted from a general-purpose DB



< 8ms

P99 read latency for compliance-indexed objects



10×

Evidence query performance vs. PostgreSQL



127

Native compliance index dimensions tracked



Zero-copy

Audit trail generation — no separate ETL pipeline required



Columnar Storage



WORM Compliance



Vector-Adjacent Index



Multi-tenancy



HIPAA-Ready

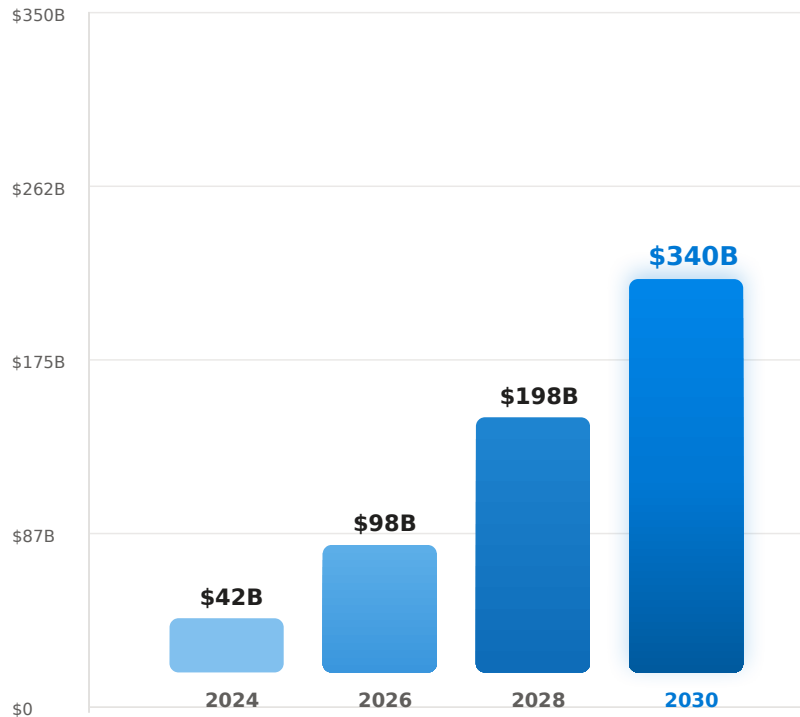


COS+ is not a database with compliance bolted on — compliance is its **primary index dimension**. Every object written carries 127 compliance attributes natively, making audit, retrieval, and governance operations first-class citizens of the query engine.

A \$340B Infrastructure Gap

The compliance-aware AI middleware market is forming now — and it is enormous

TAM GROWTH — AI COMPLIANCE INFRASTRUCTURE (USD)



Regulated Enterprises

Financial services, healthcare, legal & government requiring full auditability

\$180B



AI Platform Vendors

LLM providers & AI cloud platforms needing a compliance infrastructure layer

\$95B



System Integrators

GSI & ISV partners embedding IOS+ into vertical enterprise solutions

\$65B



TOTAL ADDRESSABLE MARKET BY 2030

\$340B

Why IOS+ Wins

No incumbent owns compliance-native AI middleware — IOS+ is building the category

COMPETITIVE COMPARISON

CAPABILITY	Legacy Middleware	Point Solutions	IOS+
Compliance Enforcement	✗	⚡	✓
Evidence Trail	✗	✗	✓
Ontology Routing	✗	⚡	✓
RAG Integration	✗	⚡	✓
Audit Speed	✗	⚡	✓
Architecture	⚡	⚡	✓
	✗	⚡	✓

DEFENSIBLE MOAT



Gate 530 Patent

FILED Q1 2026

Novel compliance-at-inference-time architecture — enforces 127 compliance dimensions inside the inference path. No competitor has filed anything comparable.



Evidence Fabric IP

PROPRIETARY PROTOCOL

Cryptographic chain-of-custody protocol linking every AI output to its source, model version, retrieval context, and gate stamp. Not replicable without the underlying ontology stack.



COS+ Schema Design

NON-RETROFITTABLE

Compliance-as-primary-index architecture baked into COS+ at the schema level. Existing general-purpose databases cannot adopt this pattern without full re-architecture.

IOS+ is the only platform where compliance is native to every layer — not added, not simulated, not retrofitted

Traction & Roadmap

IOS+ moves from architecture to revenue in four deliberate phases

PHASE 3 OF 4  IN PROGRESS

PHASE 1

COMPLETED

Q3 2025 — Foundation

- YBR-L4 Architecture finalized
- Seven-layer ontology validated
- Core Gate 530 engine built & tested

PHASE 3 — CURRENT

IN PROGRESS

Q3 2026 — Scale

- RAG Vault GA release
- Gate 530 patent filing
- Series A fundraising
- 5 enterprise pilot customers

PHASE 2

COMPLETED

Q1 2026 — First Deploy

- COS+ Database v1 deployed
- Evidence Fabric alpha released
- First pilot customer — healthcare

PHASE 4

PLANNED

Q1 2027 — Revenue

- Platform API launch
- ISV partner program
- \$10M ARR target
- SOC 2 Type II certification

ⓘ SERIES A · \$12M TARGET · MAY 2026

Join the **Infrastructure Round**

Building the **Compliance-Native AI Middleware Layer** — the trust infrastructure every enterprise needs

\$5M



ENGINEERING & PRODUCT

RAG Vault GA release · Gate 530 hardening
· COS+ v2 release · Core platform
engineering team scale-up

\$4M



GO-TO-MARKET

Enterprise sales motion · ISV partner
program launch · SOC 2 Type II certification
· 5→25 pilot customer expansion

\$3M



REGULATORY & IP

Gate 530 patent portfolio · HIPAA / GDPR
certifications · Standards body participation
· Compliance counsel

ENGINEERING & PRODUCT · 42%

GO-TO-MARKET · 33%

REGULATORY & IP · 25%

*"IOS+ is not an AI feature — it is the **infrastructure layer** that makes enterprise AI trustworthy at scale."*

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SMEPro Technologies

Houston, TX ·
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✓ HIPAA-READY

✓ SOC 2 TYPE II (TARGET Q1 2027)

✓ GDPR ALIGNED

✓ GATE 530 PATENT PENDING

✓ YBR-L4 ARCHITECTURE